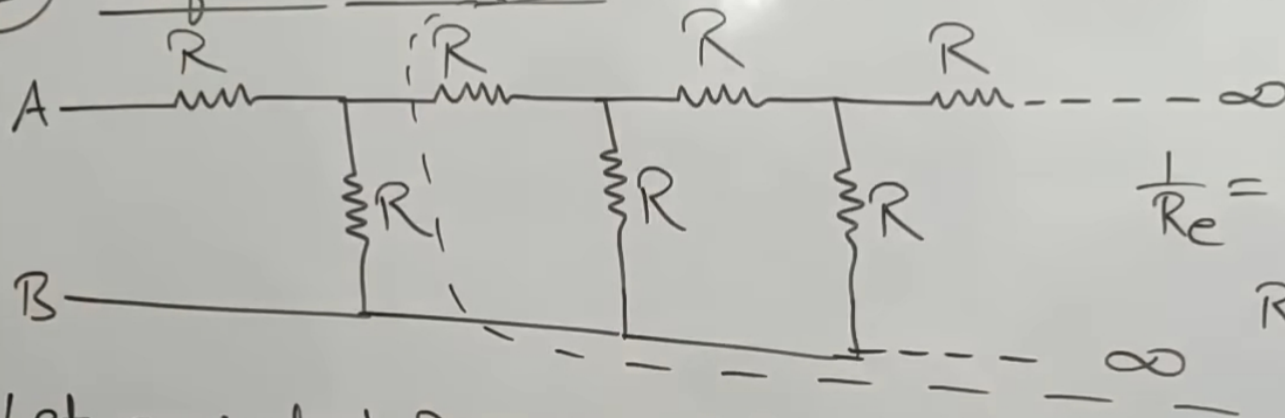


Combination of Resistors.

Find equivalent Resistance between A & B.

(II) Infinite Network.

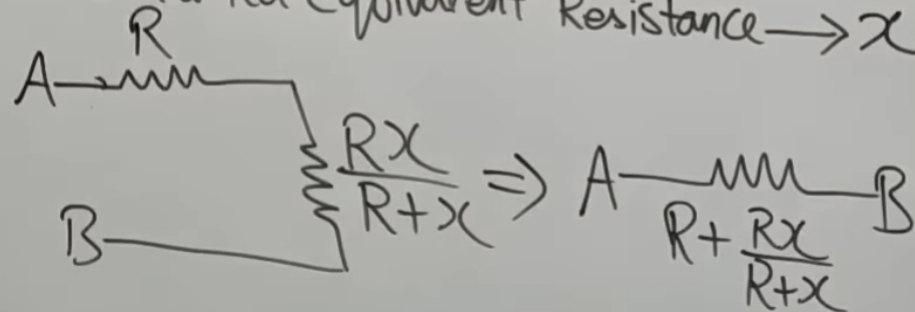
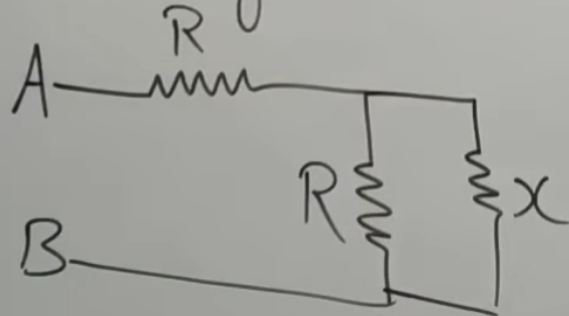


$$\frac{1}{R_e} = \frac{1}{R} + \frac{1}{x} = \frac{x+R}{Rx}$$

$$R_e = \frac{Rx}{R+x}$$

Let equivalent Resistance of this combination between A-B be 'x'.

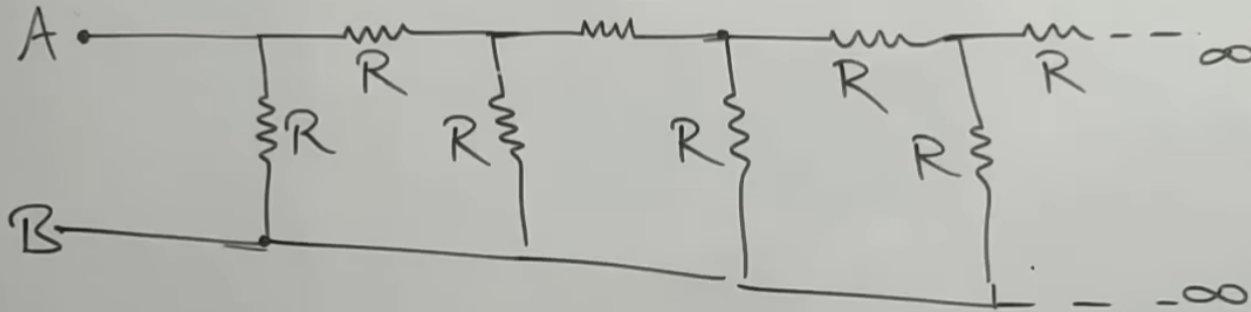
Ek Repeating unit ko chor ke baki ka equivalent Resistance $\rightarrow x$



Combination of Resistors.

II Infinite Network.

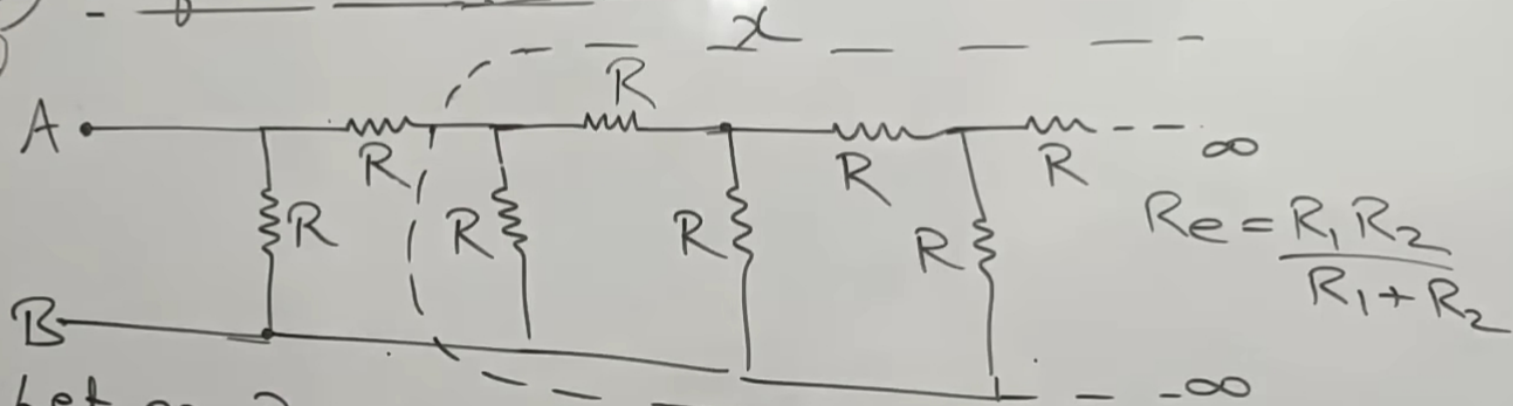
(2)



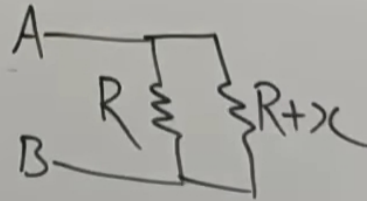
Combination of Resistors.

(II) Infinite Network.

(2)



Let eq Resistance of whole combination between A-B be ' x '
 Ek repeating unit chor k baki ka eq Resistance $\rightarrow x$

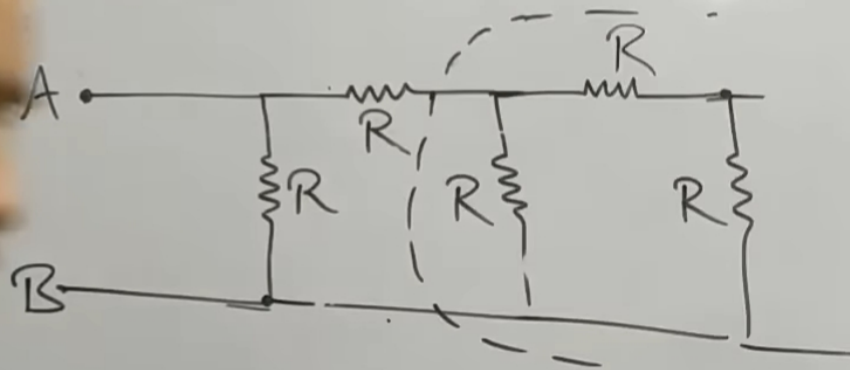


$$A \text{ --- } R \text{ --- } B$$

$$\frac{R(R+x)}{R+R+x}$$

Combination of Resistors.

1) Infinite Network.



$$X = \frac{-R \pm \sqrt{R^2 + 4R^2}}{2}$$

$$= \frac{-R \pm R\sqrt{5}}{2}$$

$$\frac{R(R+X)}{R+(R+X)} = X$$

$$R^2 + RX = X(2R + X)$$

$$R^2 + RX = 2RX + X^2$$

$$X^2 + RX - R^2 = 0$$

$$X = \frac{-R \pm \sqrt{R^2 + 4R^2}}{2}$$

$$X = \frac{-R - R\sqrt{5}}{2} \quad \text{X -ve}$$

$$X = \frac{-R + R\sqrt{5}}{2} = \frac{R(\sqrt{5} - 1)}{2} \quad \checkmark$$

Equivalent Resistance (Solved Problem 3)

Question: Find the equivalent resistance between the terminals a-b.

Infinite Ladder N/w



Solution:

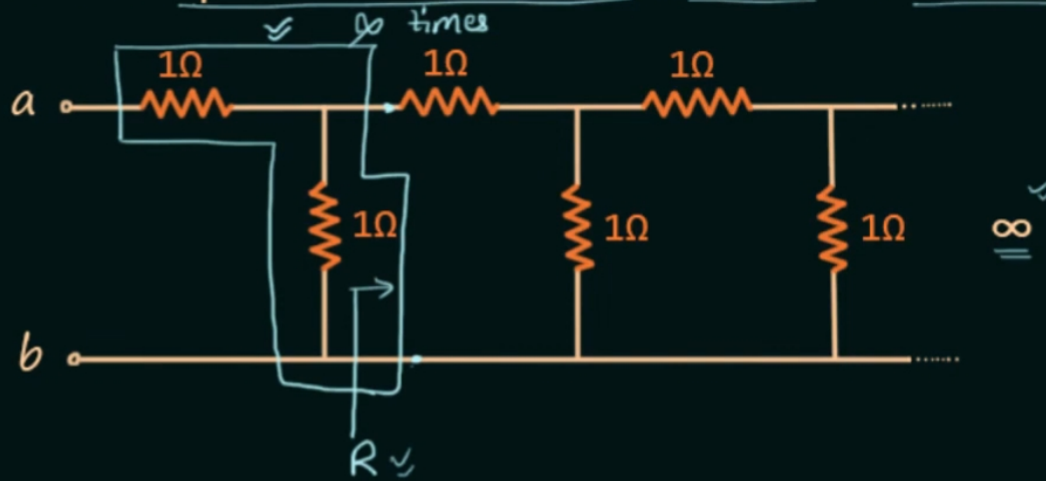


Equivalent Resistance (Solved Problem 3)

Question: Find the equivalent resistance between the terminals a-b.

Infinite Ladder N/w

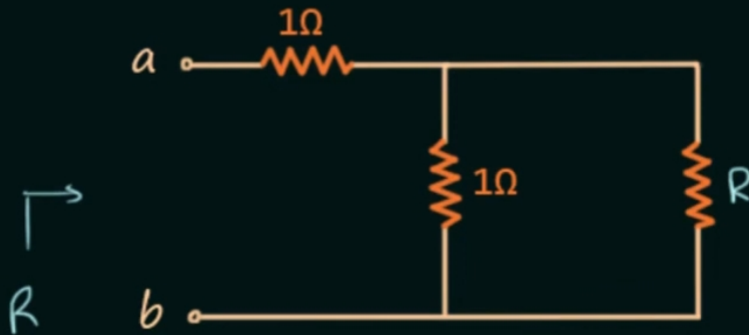
$R = ?$



$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solution:



$$R = (R \parallel 1) + 1$$

$$R = \frac{R}{R+1} + 1$$

$$R(R+1) = R + (R+1)$$

$$R^2 + R = 2R + 1$$

$$R^2 - R - 1 = 0$$

$$a=1, b=-1 \text{ \& } c=-1$$

$$R = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot (-1)}}{2 \cdot 1}$$

$$= \frac{1 \pm \sqrt{1+4}}{2}$$

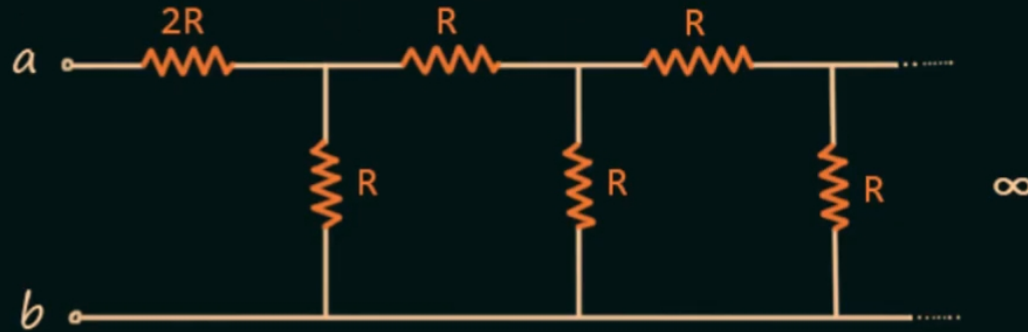
$$R = \frac{1 \pm \sqrt{5}}{2}$$

$$R = \frac{1 + \sqrt{5}}{2}, \quad \frac{1 - \sqrt{5}}{2} \quad \text{where } -R > 1$$

$$R = \frac{1 + \sqrt{5}}{2}$$



Homework: The equivalent resistance in the infinite ladder network shown in figure, is R_e



[GATE - 2014]

The value of R_e/R is _____.

An infinite ladder network of resistances is constructed with $1\ \Omega$ and $2\ \Omega$ resistances as shown in the figure. The 6 V battery between A and B has negligible internal resistance. The equivalent resistance between A and B is?

